

MA3632 Lecture 2 – Data Cleaning

In Lecture 1 we introduced the data mining pipeline and examined its opening stage in some detail: acquiring data from diverse sources and recording its provenance. We also set out Rubin’s taxonomy of missing data mechanisms – MCAR, MAR, and MNAR – and observed that the appropriate analytical response depends on which mechanism is operative. This lecture takes up the next stage of the pipeline, *data cleaning*: the detection and correction of errors, inconsistencies, and structural deficiencies before any modelling begins.

It is worth saying at the outset that data cleaning often accounts for a substantial proportion of the work in real data analysis, and that errors which survive this stage tend to propagate quietly through everything that follows. We organise the discussion around three categories of problem: missing values, outliers, and what might loosely be called structural inconsistencies. For each we look at both the mathematical content and what one actually does in practice.

1 Missing Values: Imputation

Recall from Lecture 1 that a missing value occurs whenever an entry X_{ij} is not recorded. The right response depends on the mechanism, and we now move from the taxonomy to the remedies.

1.1 Deletion

The simplest option is to discard incomplete observations.

Definition 1.1. Complete-case analysis (listwise deletion) retains only those observations for which all p variables are recorded, discarding any row with at least one missing entry.

Complete-case analysis is generally unbiased under MCAR, although validity under MAR or MNAR depends on the estimand, model, and missingness structure. Even under MCAR, discarding rows reduces the effective sample size and inflates standard errors; when missingness rates are substantial the loss of precision can be severe.

Remark 1.2. Pairwise deletion is less wasteful: each statistic is computed using all observations for which the relevant variables happen to be jointly observed. The difficulty is that the resulting covariance matrix need not be positive semi-definite, which causes problems for any method requiring matrix inversion.

1.2 Single imputation

Rather than discarding rows, one can replace each missing value with a plausible substitute. Write $X_j = (X_{1j}, \dots, X_{nj})^\top$ for the j -th variable across n observations, and let $\mathcal{O}_j \subset \{1, \dots, n\}$ be the index set of observed values.

Definition 1.3. The **mean imputation** estimator replaces each missing X_{ij} ($i \notin \mathcal{O}_j$) with

$$\hat{X}_{ij} = \bar{X}_j := \frac{1}{|\mathcal{O}_j|} \sum_{k \in \mathcal{O}_j} X_{kj}.$$

Median imputation uses the sample median of $\{X_{kj} : k \in \mathcal{O}_j\}$ in its place.

Mean imputation is computationally trivial and, under MCAR, yields an unbiased estimate of the marginal mean. Its well-known defect is that it artificially reduces the variability of X_j , since it replaces a set of observations with a single constant. To see this formally, suppose $m = |\mathcal{O}_j|$ values are observed and $n - m$ are missing, and write S_j^2 for the (Bessel-corrected) sample variance of the observed values,

$$S_j^2 = \frac{1}{m-1} \sum_{k \in \mathcal{O}_j} (X_{kj} - \bar{X}_j)^2.$$

After replacing each missing entry with $\bar{X}_j$, the numerator of the post-imputation sample variance receives no further contribution from the imputed values, since each imputed entry equals the mean exactly. The post-imputation sample variance is therefore

$$\widehat{\text{Var}}(X_j) = \frac{m-1}{n-1} S_j^2.$$

The factor $(m-1)/(n-1) < 1$ shows that variability is attenuated in proportion to the fraction of values that were missing, worsening as that fraction grows.

Remark 1.4. For skewed variables, or when outliers are present, the median may be a more robust single-imputation choice than the mean. For categorical variables the natural analogue is the mode.

1.3 Regression imputation

A more careful approach is to treat the missing variable as the response in a regression fitted on the complete cases.

Definition 1.5. In **regression imputation**, the variable X_j is modelled as

$$X_{ij} = f(\mathbf{X}_{i,-j}) + \varepsilon_i,$$

where $\mathbf{X}_{i,-j}$ denotes the remaining variables for observation i and f is some regression function. The imputed value is $\hat{X}_{ij} = \hat{f}(\mathbf{X}_{i,-j})$.

This incorporates the relationship between X_j and the remaining variables, which mean imputation does not. It still understates uncertainty and residual variability, however, since it substitutes a point prediction for a draw from a distribution.

The more complete resolution is *multiple imputation* (Rubin, 1987): $M \geq 2$ completed datasets are generated by drawing plausible values from an appropriate predictive distribution for the missing data, each is analysed separately, and the results are combined via Rubin's pooling rules. The combined standard errors reflect both the within-imputation variability and the between-imputation spread, giving properly calibrated inference. Multiple imputation is a standard principled approach to missing-data problems, though a full treatment is outside the scope of this module; pointers are given in the reading list.

Example 1.6. A housing dataset contains $n = 500$ records and $p = 6$ features, one of which – lot area – is missing for 40 properties (8%). Examination of the missingness pattern shows that all 40 gaps occur in properties built before 1970. This is consistent with an MAR mechanism: the probability of a missing lot area depends on the observed variable *year built*, not on the lot area itself.

Complete-case analysis would remove the 40 older properties and could bias subsequent analyses by systematically under-representing older properties. Regression imputation of lot area on year built and the remaining features is better motivated: it exploits the systematic relationship between age and plot size to produce plausible substitutes while retaining the full sample of 500 observations.

2 Outliers

An outlier is an observation that departs markedly from the bulk of the data. The term is intentionally informal: what constitutes a departure depends on the assumed model, the variable type, and the purpose of the analysis. We distinguish *univariate* outliers (anomalous values of a single variable) from *multivariate* outliers (observations that are unusual in the joint distribution even though each coordinate may look unremarkable in isolation).

2.1 Univariate detection

Definition 2.1. The **interquartile range** of a sample $x_1, \dots, x_n$ is $\text{IQR} = Q_3 - Q_1$. The **Tukey fence criterion** flags x_i as a (mild) outlier when

$$x_i < Q_1 - 1.5 \text{IQR} \quad \text{or} \quad x_i > Q_3 + 1.5 \text{IQR},$$

and as an extreme outlier when the coefficient 1.5 is replaced by 3.0.

The Tukey fence makes no distributional assumption and is straightforward to compute. Its limitation is that it operates one variable at a time and will miss outliers whose extremity lies in a combination of features rather than in any single one.

A parametric alternative assumes approximate normality and uses the z -score.

Definition 2.2. The z -score of x_i is

$$z_i = \frac{x_i - \bar{x}}{s},$$

where $\bar{x}$ and s are the sample mean and standard deviation. Under normality, approximately 99.7% of observations lie within three standard deviations of the mean; the threshold $|z_i| > 3$ flags observations falling outside this range.

Remark 2.3. The z -score has a well-known weakness: a single extreme value inflates both $\bar{x}$ and s , which can reduce the apparent z -score of the very observation one is trying to flag. This phenomenon is called *masking*. A robust alternative is to replace $\bar{x}$ and s with the sample median and the *median absolute deviation*, $\text{MAD} = \text{median}(|x_i - \text{median}(x)|)$.

2.2 Multivariate detection

Definition 2.4. Let $\hat{\boldsymbol{\mu}} \in \mathbb{R}^p$ and $\hat{\boldsymbol{\Sigma}} \in \mathbb{R}^{p \times p}$ be the sample mean vector and covariance matrix. The **Mahalanobis distance** of observation $\mathbf{x}_i \in \mathbb{R}^p$ is

$$D_i = \sqrt{(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{x}_i - \hat{\boldsymbol{\mu}})}.$$

Under multivariate normality, a χ_p^2 distribution is commonly used as an approximate reference distribution for D_i^2 , particularly in large samples; for example, observations for which D_i^2 exceeds the upper 2.5% point of χ_p^2 may be flagged for further scrutiny. Mahalanobis distance accounts for the correlation structure of the data and identifies observations that are unusual in their joint configuration, something coordinate-wise methods cannot do.

Example 2.5. Consider a dataset recording height and weight for $n = 200$ individuals. An individual with height 195 cm and weight 55 kg has a z -score of roughly 1.8 for height and -1.6 for weight, neither of which crosses a standard univariate threshold. Yet this combination – unusually tall and unusually light – sits far from the main cluster in the joint distribution and is correctly identified as anomalous by the Mahalanobis distance.

2.3 What to do with outliers

Detection is only the first step; one must then decide how to treat a flagged observation. The available options are the following.

Investigate and correct. The outlier may simply be a recording error – a weight of 550 kg instead of 55 kg, say – that can be corrected by consulting the source. This should always be the first question.

Retain. If the value is genuine and the analysis is meant to describe the full population, removal is not justified. Robust estimators are the appropriate response.

Winsorise. Replace extreme values beyond specified cut-offs with the corresponding boundary values, thereby limiting their influence while retaining the observations.

Remove. Deletion is appropriate only when an observation is demonstrably erroneous and cannot be corrected, or when the analysis is explicitly restricted to a subpopulation that excludes such values. Any removal must be recorded and justified.

Remark 2.6. The decision to remove an outlier is a scientific judgement, not a statistical one. A large z -score or Mahalanobis distance is evidence of unusualness, not of error. Domain knowledge must inform the final decision.

3 Structural Inconsistencies

Beyond missing values and outliers, real datasets frequently contain defects of a more mundane kind. We survey the most common.

3.1 Duplicate records

A duplicate is a record that appears more than once, either as an exact copy or as a near-duplicate referring to the same entity. Exact duplicates arise from pipeline errors or repeated imports. Near-duplicates – the same customer recorded under two slightly different name spellings, for instance – are harder to detect and require *entity resolution*.

Definition 3.1. Entity resolution (record linkage, deduplication) is the problem of partitioning a set of records $\mathcal{R} = \{r_1, \dots, r_n\}$ into equivalence classes such that every record in a class refers to the same real-world entity.

Exact matching on a key field is the simplest case. Probabilistic methods use similarity measures – edit distance for strings, absolute difference for numerics – and a threshold, often combined with blocking to avoid the $O(n^2)$ cost of comparing every pair.

3.2 Type and format errors

Variables are frequently stored in inconsistent or incorrect formats: dates as free text ("01/06/2024" alongside "2024-06-01"), categorical variables encoded with inconsistent capitalisation ("Male", "male", "M"), or numeric columns stored as strings because of embedded currency symbols. These errors are detectable by inspecting the inferred dtype of each column together with a sample of its values. They must be resolved before any quantitative operation; attempting arithmetic on a column of type `object` will either raise an error or, worse, silently produce nonsense.

3.3 Constraint violations

Many variables have well-defined domains: an age cannot be negative, a probability cannot exceed one, a postal code must match a known pattern. Observations that violate these constraints are either erroneous or reflect some coding convention that needs to be understood before the data can be used.

Definition 3.2. A **constraint violation** is an observation x_i for which $x_i \notin \mathcal{D}$, where $\mathcal{D}$ is the domain of the variable. A **referential integrity violation** in a relational dataset occurs when a foreign key references a primary key that does not exist in the referenced table.

It is good practice to encode domain constraints in an explicit validation schema – using a tool such as the Python **pandera** library, for example – so that violations are caught automatically whenever new data are ingested.

4 A Cleaning Workflow

The problems discussed above are not neatly separable, and the order in which they are addressed matters. The following sequence is standard in practice.

- (1) *Audit.* Before touching anything, compute per-column missing rates, dtypes, unique value counts, and basic descriptive statistics. Understand the scale of the problem first.
- (2) *Structural repair.* Fix type errors, format inconsistencies, and constraint violations. These are generally deterministic corrections with little ambiguity.
- (3) *Deduplication.* Remove or merge duplicate records before imputing, since duplicates distort the estimates used in imputation.
- (4) *Outlier investigation.* Apply detection methods, examine flagged observations, and document decisions. Do not remove values without first checking whether they are genuine.
- (5) *Missing-value treatment.* Select an appropriate strategy informed by the observed missingness pattern, subject-matter knowledge, and assumptions about the missingness mechanism, and document the treatment applied.
- (6) *Re-audit.* Run the same diagnostics as in step (1) and verify that the defect counts have fallen as expected.

Remark 4.1. The entire cleaning pipeline should be written as a reproducible script that operates on the raw data without modifying it. If the raw data are updated, or if the analysis needs to be rerun, every decision must be recoverable. A dataset whose cleaning history is undocumented is a dataset that cannot be trusted.

5 Exercises

Exercise 5.1 (Applied). A dataset of $n = 1\,000$ patient records contains a variable `systolic_bp` (systolic blood pressure, in mmHg). An audit reveals: 47 values are missing; a further 3 values are recorded as 0; the sample mean of the remaining 950 values is $\bar{x} = 122.4$ with sample standard deviation $s = 18.7$; and clinical records show that all 47 missing values belong to patients admitted for elective procedures, who tend as a group to have lower blood pressure than emergency admissions.

- (a) Classify the three zero-valued records and describe how you would treat them.
- (b) What is the most plausible missing data mechanism for the 47 missing values? Justify your answer.
- (c) A colleague proposes mean imputation. Give two distinct reasons why this is unsuitable here, and suggest a better approach.
- (d) The Tukey fence is applied to the 950 valid observations. Given $Q_1 = 109$ and $Q_3 = 134$, compute the upper fence and estimate how many observations you would expect to exceed it under normality.

Solution. (a) A systolic blood pressure of zero is physiologically impossible and constitutes a constraint violation. These three records should be recoded as missing (NaN) and treated alongside the other 47; the corrected missing count is therefore 50.

- (b) The most plausible mechanism is MAR. Whether a blood pressure value is missing depends on admission type (elective vs. emergency), which is itself observed. It is not plausibly MCAR because missingness is concentrated in one admission group rather than occurring at random across the full sample. There is no evidence from the data that it is MNAR, since there is no suggestion that patients with particularly low or high blood pressure are systematically more likely to have missing values; however, as with any observational audit, MNAR cannot be definitively ruled out.
- (c) First, the observed values over-represent emergency patients, who have higher average blood pressure, so $\bar{x} = 122.4$ is an upwardly biased estimate of the mean for elective patients; imputing with this value inflates the blood pressure of all 50 missing records. Second, replacing 50 distinct observations with the single constant 122.4 artificially deflates the variance of the variable, distorting its covariance and correlation relationships with other variables and understating uncertainty in any subsequent model. Regression imputation (or multiple imputation) using admission type as a predictor is more appropriate, since it exploits the known relationship between admission type and blood pressure.
- (d) $IQR = 134 - 109 = 25$, so the upper fence is $134 + 1.5 \times 25 = 171.5$ mmHg. To estimate the expected exceedance count under normality, note that $(171.5 - 122.4)/18.7 \approx 2.63$, so the fence lies approximately 2.63 standard deviations above the mean. Since $P(Z > 2.63) \approx 0.0043$, the expected number of observations above the fence is approximately $950 \times 0.0043 \approx 4$.

□

Exercise 5.2 (Mathematical). Let $x_1, \dots, x_n$ be a random sample from a distribution with mean μ and variance σ^2 . Suppose $m < n$ values are observed and the remaining $n - m$ are missing completely at random. Write $\hat{\mu}_{\text{obs}}$ for the sample mean of the m observed values, and $\hat{\mu}_{\text{imp}}$ for the mean of the full n values after replacing each missing entry by $\hat{\mu}_{\text{obs}}$.

- (a) Show that $\hat{\mu}_{\text{imp}} = \hat{\mu}_{\text{obs}}$.
- (b) Let S_{obs}^2 be the sample variance of the m observed values. Compute the sample variance S_{imp}^2 after mean imputation and express it in terms of S_{obs}^2 , m , and n .
- (c) Interpret the result of (b). When does $S_{\text{imp}}^2 = S_{\text{obs}}^2$?

Solution. (a) After imputation the n values are $x_1, \dots, x_m$ together with $n-m$ copies of $\hat{\mu}_{\text{obs}}$. Their sum is

$$\sum_{i=1}^m x_i + (n-m)\hat{\mu}_{\text{obs}} = m\hat{\mu}_{\text{obs}} + (n-m)\hat{\mu}_{\text{obs}} = n\hat{\mu}_{\text{obs}},$$

so $\hat{\mu}_{\text{imp}} = \hat{\mu}_{\text{obs}}$.

(b) Since $\hat{\mu}_{\text{imp}} = \hat{\mu}_{\text{obs}}$ by part (a), the numerator of S_{imp}^2 is

$$\begin{aligned} \sum_{i=1}^n (x_i - \hat{\mu}_{\text{imp}})^2 &= \sum_{i=1}^m (x_i - \hat{\mu}_{\text{obs}})^2 + \sum_{j=1}^{n-m} (\hat{\mu}_{\text{obs}} - \hat{\mu}_{\text{obs}})^2 \\ &= (m-1) S_{\text{obs}}^2, \end{aligned}$$

and therefore

$$S_{\text{imp}}^2 = \frac{(m-1) S_{\text{obs}}^2}{n-1}.$$

(c) Since $m \leq n$, the ratio $(m-1)/(n-1) \leq 1$, so mean imputation always deflates the sample variance. The deflation worsens as the proportion of missing values $1-m/n$ increases. The two variances coincide only when $m = n$, that is, when no imputation has been performed. $\square$

Further Reading

Rubin (1987) gives the original treatment of multiple imputation and the pooling rules used to combine estimates across completed datasets. Little and Rubin (2019) is a standard reference for the missing-data mechanisms introduced in Lecture 1 and used throughout this lecture. Bantilan (2020) introduces **pandera** as a framework for statistical data validation of the kind discussed in Section 3.3.

References

- Bantilan, N., 2020. **pandera**: Statistical Data Validation of Pandas Dataframes. In *SciPy* (pp. 116–124).
- Little, R.J.A. and Rubin, D.B., 2019. *Statistical Analysis with Missing Data*. 3rd ed. John Wiley & Sons.
- Rubin, D.B., 1987. *Multiple Imputation for Nonresponse in Surveys*. New York: John Wiley & Sons.